

Label-Efficient Satellite Land Cover Classification via Contrastive Self-Supervised Representation Learning

Team No. 15

Nishanth (AM.SC.U4AIE24025) — Kalidasan B (AM.SC.U4AIE24025)

Deep Learning Hackathon — Track 4 — Dataset: EuroSAT (RGB)

September 2026

1 Problem

Labeled satellite imagery is scarce and expensive, while unlabeled imagery is abundant. Earth observation satellites now produce far more imagery than can be manually annotated, and land-cover classification – assigning each image to a category such as forest, residential, or river – underpins applications in agriculture, urban planning, and climate monitoring that are bottlenecked by this labeling cost. This report tests whether self-supervised contrastive pretraining reduces the labeled-data requirement for EuroSAT land-cover classification. Specifically: *does SimCLR pretraining improve accuracy over a supervised-from-scratch baseline when only 1% or 10% of labels are available, and does the benefit persist under fine-tuning as well as linear probing?*

2 Related Work

Convolutional architectures pretrained on natural images (e.g. ImageNet) have been widely repurposed for satellite scene classification via transfer learning, but the domain gap between natural and overhead imagery limits how well such features transfer, particularly for fine-grained land-cover distinctions. EuroSAT was introduced specifically to benchmark land-use classification from Sentinel-2 imagery[2] and has become a standard testbed given its modest image size and balanced classes, though prior work has mostly reported fully supervised accuracy with the entire labeled set available – leaving the low-label regime this report targets comparatively underexplored.

Contrastive self-supervised methods learn representations by pulling together embeddings of augmented views of the same image while pushing apart embeddings of different images, without requiring labels. SimCLR showed that this simple framework, combined with a learned nonlinear projection head and a large batch size, can produce representations that rival supervised pretraining once a linear classifier is trained on top[1]. Momentum-based variants such as MoCo relax the large-batch requirement via a queue of negatives, while non-contrastive methods (BYOL, SimSiam) remove the need for explicit negatives entirely. We adopt SimCLR here for its simplicity and well-understood loss, making it a natural first baseline before considering these alternatives.

3 Method

Dataset. EuroSAT RGB: 27,000 labeled 64×64 Sentinel-2 patches, 10 land-cover classes (e.g. Forest, Highway, Residential, River, SeaLake). A fixed test split is held out; stratified 1% and 10% label subsets are sampled per class for downstream training, keeping both budgets class-balanced and directly comparable across methods.

Pretraining. A ResNet-18[3] encoder (f , trained from scratch, 512-d output) is pretrained with SimCLR on the full unlabeled pool. Two augmented views per image (random resized crop, horizontal/vertical flip, color jitter, random grayscale; Gaussian blur is omitted given EuroSAT’s low native resolution) are passed through f and a projection head g ($512 \rightarrow 512 \rightarrow 128$, ReLU between layers), and optimized with the NT-Xent loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad z_i = g(f(\tilde{x}_i)), \quad (1)$$

where $\text{sim}(u, v) = u^\top v / (\|u\| \|v\|)$ is cosine similarity. The projection head is discarded after pretraining; only f is kept for downstream use.

Downstream evaluation. At each label budget (1%, 10%): (i) *Linear probe* – freeze f , train a linear classifier on its pooled features; (ii) *Fine-tuning* – unfreeze f , jointly train with the classifier at a lower encoder learning rate to avoid destroying the pretrained representation early on; (iii) *Supervised baseline* – train the same ResNet-18 from random initialization on only the labeled subset, isolating the effect of pretraining from

architecture/optimizer choices. Downstream runs reported here use a single seed (42); averaging over multiple seeds is left for future work.

Table 1: Training configuration by stage.

Stage	Epochs	Batch Size	Optimizer
SimCLR Pretraining	100	128	LARS
Linear Probe	50	64	SGD
Fine-Tuning	75	64	Adam
Supervised Baseline	75	64	Adam

SimCLR temperature $\tau = 0.5$; pretraining LR 3×10^{-4} with cosine decay and 10-epoch warmup, weight decay 1×10^{-6} . Fine-tuning encoder LR 1×10^{-4} ; baseline/probe LR 1×10^{-3} with step decay; augmentation limited to random crop and flip to avoid conflating augmentation strength with pretraining benefit.

4 Results

Table 2: Downstream test accuracy at fixed label budgets (single seed 42). Baseline = supervised from scratch.

Method	Acc@1% (270 labels)	Acc@10% (2700 labels)
Supervised From Scratch (baseline)	64.22	81.74
SimCLR + Linear Probe	76.63	82.44
SimCLR + Fine-Tuning	78.59	87.33

SimCLR pretraining (100 epochs, NT-Xent train loss $4.519 \rightarrow 3.731$, label-free validation pair-retrieval accuracy reaching 0.893 at epoch 97) improves downstream accuracy over the from-scratch baseline at both budgets: **+14.4pp at 1%** (78.59% vs. 64.22%) and **+5.6pp at 10%** (87.33% vs. 81.74%), confirming the hypothesis that the SSL benefit is largest where labeled signal is scarcest. Fine-tuning outperforms the frozen linear probe at both budgets (+2.0pp at 1%, +4.9pp at 10%), with the gap widening as more labels become available to adapt the backbone. Per-class recall shows Highway and PermanentCrop are the hardest categories for every method at both budgets, while fine-tuning’s largest gains over the baseline concentrate in River (+32pp @ 1%), Pasture (+30pp @ 1%), and HerbaceousVegetation (+29pp @ 1%) – classes whose visual similarity to neighboring categories likely benefits most from SimCLR’s learned invariances.

5 Ablation: Effect of Self-Supervised Pretraining

Table 3: Ablation at the 10% label budget, holding architecture and training procedure fixed.

Configuration	Accuracy
Supervised baseline (no SSL)	81.74
SSL pretraining + Linear Probe	82.44
SSL pretraining + Fine-Tuning	87.33

This isolates the marginal contribution of SimCLR pretraining alone: the probe row tests whether frozen SSL features are directly linearly separable, and the fine-tuning row tests whether SSL still provides a better initialization once the backbone is allowed to adapt. At 10%, the linear probe barely edges out the baseline (+0.7pp), meaning frozen SimCLR features are only marginally more linearly separable than what the baseline learns from 2,700 labels alone. Fine-tuning, however, pulls well ahead of both (+5.6pp over baseline, +4.9pp over the probe), indicating that the encoder’s value at this budget lies mainly in providing a better initialization for adaptation, not in producing an already-linear feature space.

6 Discussion

Together, the main results and the ablation explain *why* the accuracy gain occurs, not just that it does. As expected, the SSL benefit is largest at the 1% budget (+14.4pp), where a from-scratch model has too little signal to learn discriminative filters while the SimCLR encoder has already learned general texture statistics from the unlabeled pool; at 10%, the supervised baseline partially closes this gap, shrinking the advantage to +5.6pp, consistent with EuroSAT’s relatively low visual complexity letting 2,700 labels go a

long way on their own. Comparing the linear probe and fine-tuning rows against the same baseline further shows that the value of pretraining shifts with label budget: at 1%, the frozen probe already captures most of the benefit (76.63% vs. 78.59% fine-tuned), so an already-linear feature space matters most when labels are scarcest; at 10%, the probe’s edge over the baseline nearly vanishes while fine-tuning pulls ahead, meaning the pretrained encoder’s role shifts toward providing a better initialization once enough labels are available to justify adapting the backbone. The persistent difficulty of Highway and PermanentCrop across all methods and budgets suggests these classes’ errors stem from visual overlap with related categories (e.g. Highway with PermanentCrop/Industrial) rather than from label scarcity alone, since SSL pretraining narrows but does not close their recall gap.

7 Limitations

- Evaluation restricted to EuroSAT RGB; may not transfer to other sensors, resolutions, or multispectral bands.
- SimCLR is sensitive to augmentation choice; the pipeline here was tuned qualitatively for EuroSAT’s low resolution, not exhaustively searched.
- SimCLR pretraining took roughly 51 minutes on a single L40S GPU (2.45GB peak memory, ~ 1008 samples/s) for 100 epochs, a wall-clock and memory cost not amortized anywhere in the accuracy table.
- A handful of pretraining steps hit non-finite gradients under mixed precision and were skipped rather than applied, slightly reducing the effective number of gradient updates during Stage 1.
- Each configuration is evaluated with a single seed (42); the reported accuracies do not reflect run-to-run variance.
- Label budgets are evaluated only at 1% and 10%; the full label-efficiency curve (e.g. 2%, 5%, 20%) is not characterized.
- No comparison to other SSL objectives (MoCo, BYOL) or to ImageNet-pretrained initialization, both of which are natural extensions.

8 Conclusion

This report evaluates label-efficient land-cover classification on EuroSAT using SimCLR-based self-supervised pretraining, comparing linear probing and fine-tuning of a pretrained ResNet-18 encoder against a supervised-from-scratch baseline under matched 1% and 10% label budgets. The controlled comparison isolates the contribution of self-supervised pretraining itself: SimCLR pretraining improves accuracy over the from-scratch baseline at both budgets (+14.4pp @ 1%, +5.6pp @ 10%), with fine-tuning outperforming the frozen linear probe throughout. The 10% ablation shows this value shifts with label budget – a mostly linear-separability benefit at 1%, and mostly a better-initialization benefit at 10%, where the frozen probe’s edge over the baseline nearly disappears. Future work could extend this comparison to additional SSL objectives (MoCo, BYOL), multispectral EuroSAT bands, multi-seed variance estimates, and a denser sweep of label budgets to characterize the full label-efficiency curve rather than two fixed points.

References

- [1] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A Simple Framework for Contrastive Learning of Visual Representations,” *ICML*, 2020.
- [2] P. Helber, B. Bischke, A. Dengel, and D. Borth, “EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification,” *IEEE JSTARS*, vol. 12, no. 7, pp. 2217–2226, 2019.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” *CVPR*, 2016.